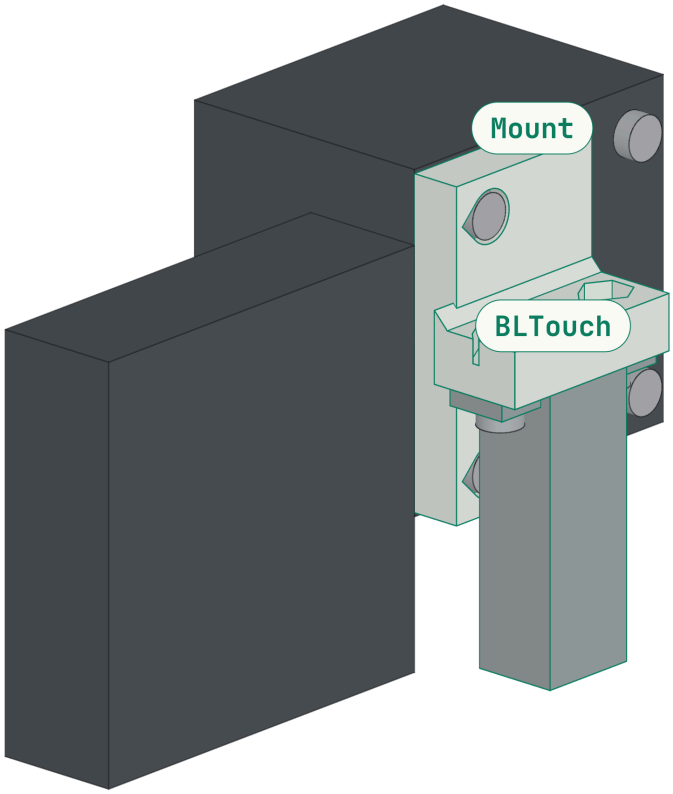
LULZBOT TAZ WORKHORSE · SE TOOL HEAD · BLTOUCH

BLTouch mount

A printed mount that holds the BLTouch beside the front of the extruder motor. It screws on with the two front motor screws, so it stays put when the head gets warm. The pin is close to the nozzle in Y, so the probe reaches the full bed.



The probe offsets

- x_offset 54.7 mm, y_offset -17.1 mm (the old bracket: 12, -76)
- The pin reaches X 4.7 to 347.7 and Y -34.1 to 290.9. The bed is 0 to 280.
- The bottom of the BLTouch is 3.3 mm over the nozzle tip (antclabs: 2.3 to 4.3 mm).
- At X 293 the X end passes 2.0 mm behind the mount.

Estimates

The motor, the X end and the BLTouch values are measured. The place of the blower fan, the ear thickness and the screw heads are estimates. MEASURE.md lists them. The mount itself is the fit test: it is small.

Parts and settings

Qty	Part	Print size	PETG
1	**bltouch mount** Back face on the bed, the front up. Solid. No supports. It is also the fit test.	18 × 41 × 26 mm	12 g
2	**M3 screws, 4 mm longer than the front motor screws** They replace the two front screws of the motor. The plate is 4 mm thick under each head.		
2	**M3 × 10 screws** From under the BLTouch ears, up through the shelf.		
2	**M3 nuts** In the pockets on top of the shelf.		

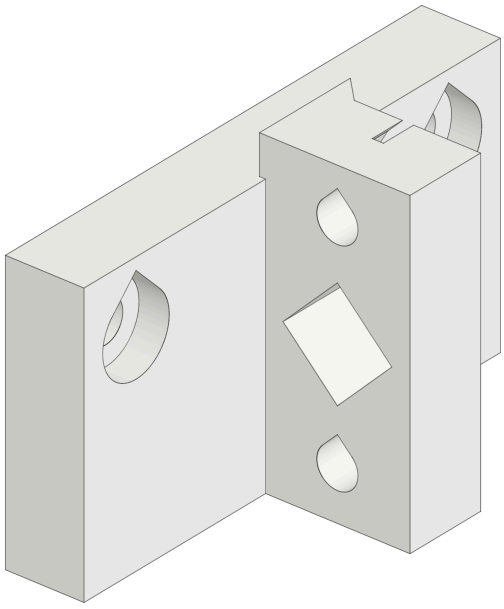
Filament: about 12 g of PETG, $0.28 at $22.99/kg.

Print

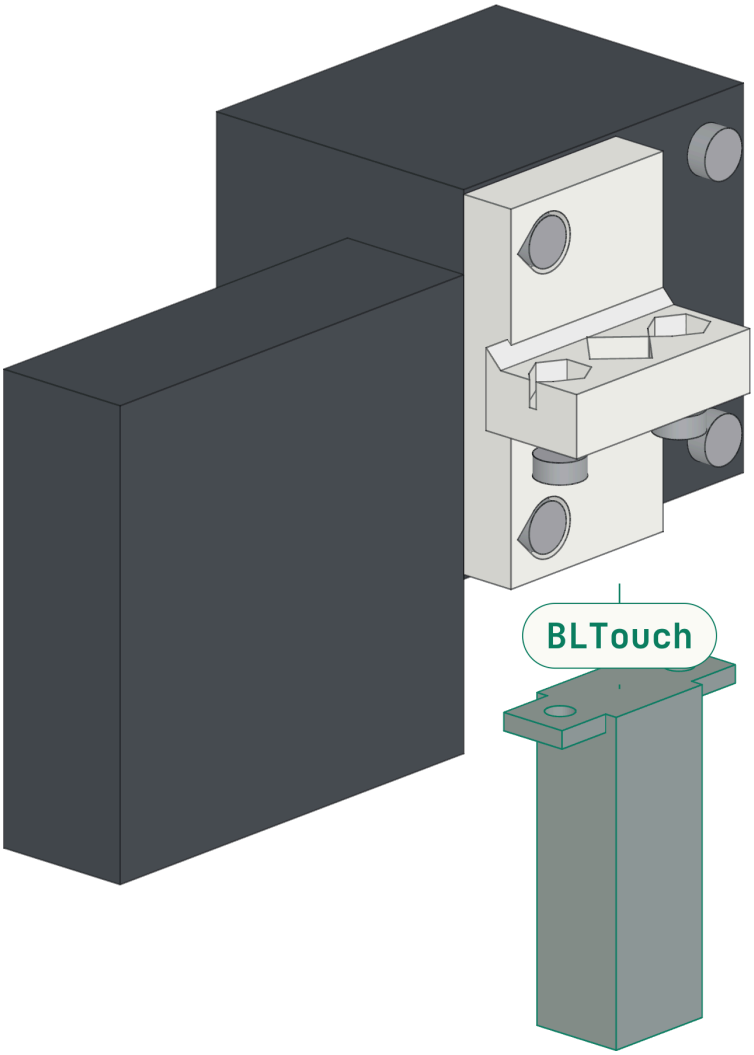
- PETG, 2.85 mm, 0.5 mm nozzle. PLA gets soft next to a warm motor.
- 240 °C nozzle, 75 °C bed. These are starting values for this printer.
- 0.2 mm layers, 2 mm walls, 100 % infill, skirt. No supports.
- Profiles: `python3 cura/make_profiles.py --install` with Cura closed, then pick “BLTouch mount - …”.

Tools

- Phillips screwdriver for the motor screws.
- 2.5 mm hex key or screwdriver for the BLTouch screws, small pliers for the nuts.

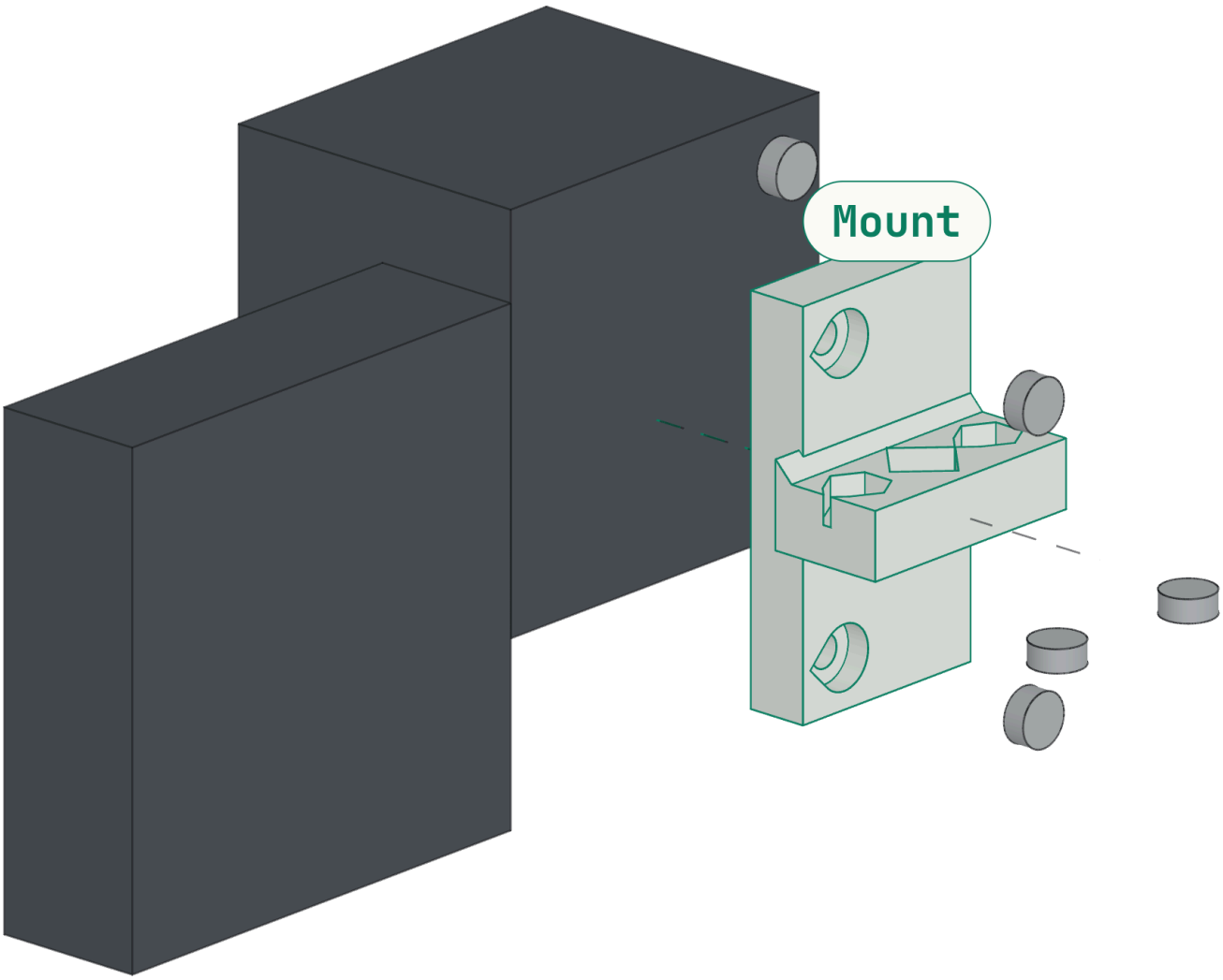


Step 1: Put the BLTouch on the mount



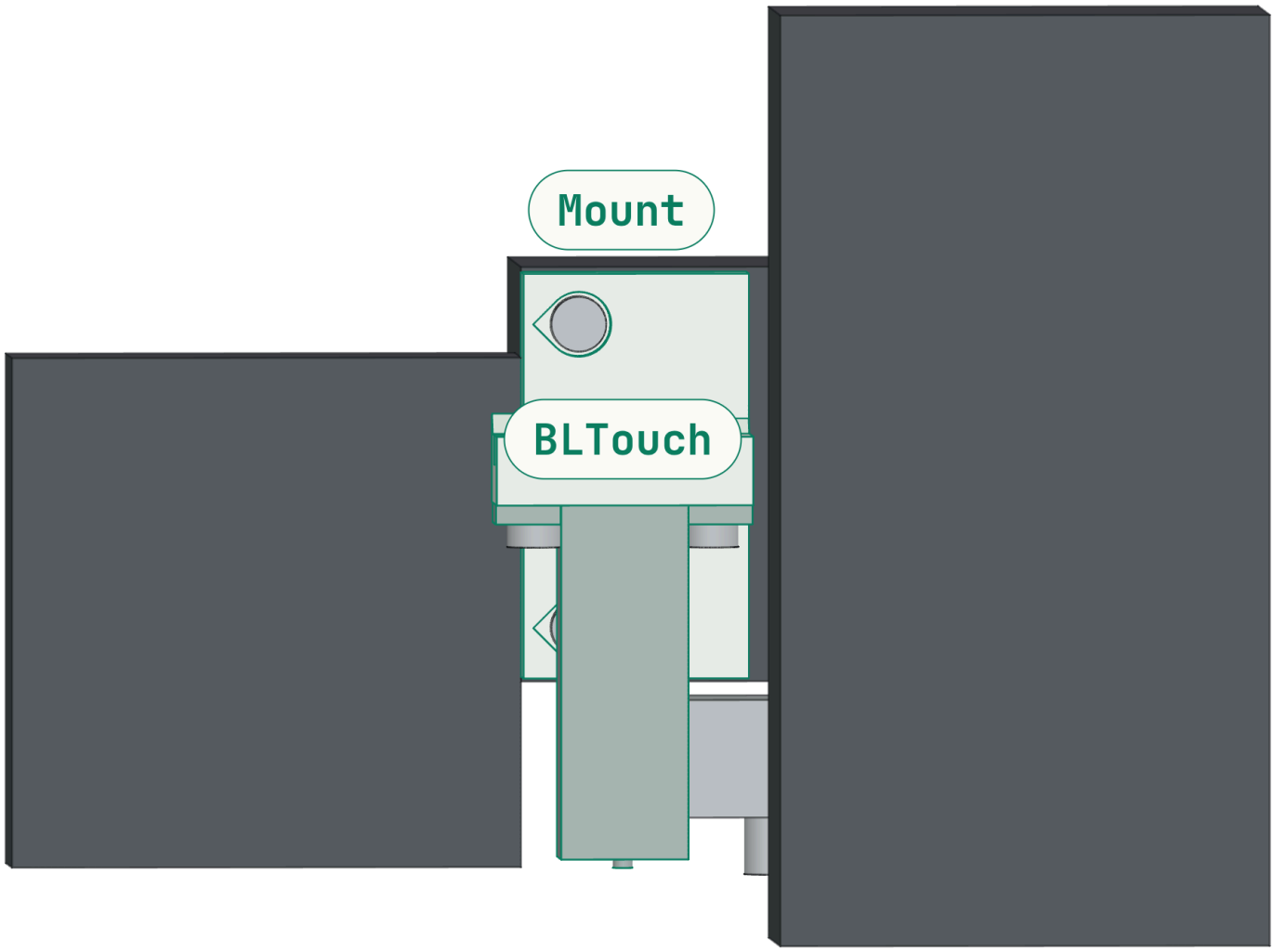
1. Turn off the printer. Remove the BLTouch and the old L bracket from the ear of the blower fan. Keep the BLTouch plugged in.
2. Put an M3 nut in each pocket on top of the shelf.
3. Feed the BLTouch wires up through the diamond hole in the shelf.
4. Hold the ears against the bottom of the shelf. Put in the two M3 × 10 screws from below. Tighten them.

Step 2: Change the front motor screws



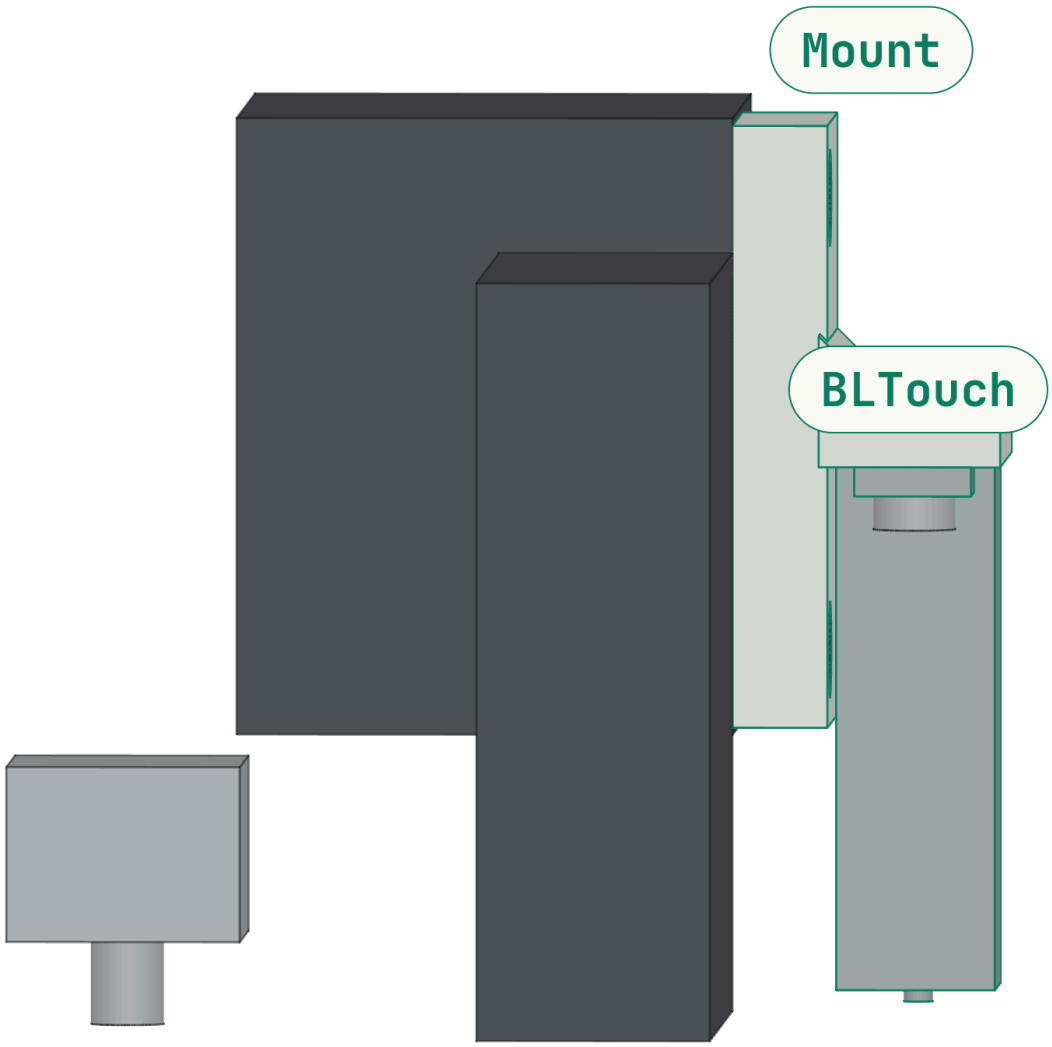
1. Remove only the upper front motor screw. Measure its length. The new screw is 4 mm longer.
2. Hold the mount on the motor face, the BLTouch down and to the front. Put the new screw through the upper hole and tighten it a little.
3. Do the same with the lower front screw. Then tighten both. Change one screw at a time, so the motor stays in place.

Step 3: Check the clearance



1. Move the head by hand to the right end of X. The X end must pass behind the BLTouch.
2. Look from the front: the bottom of the BLTouch must be about 3.3 mm above the nozzle tip.
3. Route the wires along the old path. Tie them clear of the X end.

Step 4: Set the printer



1. In printer.cfg, set x_offset: 54.7 and y_offset: -17.1 in [bltouch].
2. Set the mesh to the full bed: mesh_min 15, 15 and mesh_max 265, 265.
3. Set the z_tilt points so the pin is at X 15 and X 265.
4. Do the paper test for z_offset, and the axis twist calibration again. Then probe the full bed hot.

Adjust the fit

All the values are in `shape.py`. Change a value, then run `./make.sh`. The fit check stops the build if a part touches the motor, the blower, the X end or the hot end.

The BLTouch is too high or too low. Measure from the nozzle tip to the bottom of the BLTouch body, pin up. Set `BLT_GAP` (2.3 to 4.3).

The X end touches the BLTouch. Measure how far the X end starts behind the front of the motor. Set `X_END_Y0`.

The screw holes do not line up. Measure from the bottom of the motor and from the front of the motor to the centre of each front screw. Set `FRONT_SCREWS`.

The plate touches the blower fan. Set `BLOWER` from a measurement.

The nuts are loose in the pockets. Set `NUT_AF` 0.2 mm smaller.